OTH REGENSBURG

Praktikum Regelungstechnik (RTP)

Versuch T2: Systemidentifikation

Versuch 1

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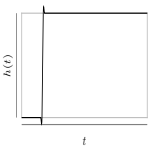
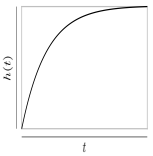
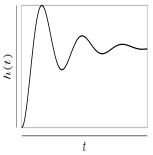
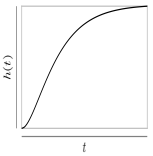
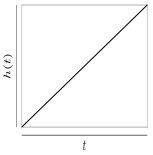
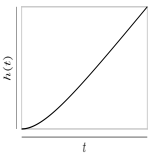
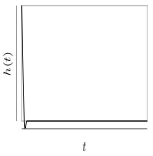
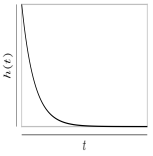
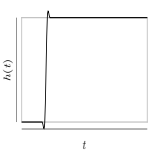
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1 Preperations

In preparation for this experiment, the files provided in ELO were downloaded and saved. In addition, the step responses for various system behaviours were researched. As tasked in Tasks $task:a)V$ and $task:b)V$ $task:c)V$ $task:d)V$

1.1. Different basic system behaviors

Systemverhalten	Übertragungsfunktion	Visualization
P-Glied (K_P)	$G(s) = K_P$ (1.1)	
PT_1-Glied (K_P, T_1)	$G(s) = \frac{K_P}{T_1 s + 1}$ (1.2)	
PT_2-Glied (K_P, ω_0, D)	$G(s) = \frac{K_P}{\frac{1}{\omega_0^2}s^2 + \frac{2D}{\omega_0}s + 1}$ (1.3)	
PT_2-Glied für $D \geq 1$ (K_P, T_1, T_2)	$G(s) = \frac{K_P}{(T_1 s + 1)(T_2 s + 1)}$ (1.4)	
I-Glied (K_I)	$G(s) = \frac{K_I}{s}$ (1.5)	
IT_1-Glied (K_I, T_1)	$G(s) = \frac{K_I}{s(T_1 s + 1)}$ (1.6)	
D-Glied (K_D)	$G(s) = K_D\, s$ (1.7)	
DT_1-Glied (K_D, T_1)	$G(s) = \frac{K_D\, s}{T_1 s + 1}$ (1.8)	
T_t-Glied (K_P, T_t)	$G(s) = K_P\, e^{-T_t s}$ (1.9)	

1.1.1. P-Glied

Explanation

Proportional-Glied, show a constant behavior in steady state for a constant input. The output is directly proportional to the input, with the proportionality constant being K_P. The equation for a P-Glied is given by

$$x_a(t) = K_P x_e(t) \tag{1.10}$$

where $x_a(t)$ is the output and $x_e(t)$ is the input. The transfer function of a P-Glied is $G(s) = K_P$ 1.1.

Parameters needed for characterization

The only parameter needed to characterize a P-Glied is the proportional gain K_P. This parameter determines the steady-state gain of the system and how much the output will change in response to a change in the input.

Step and Impulse response

The step response of a P-Glied is a step function that reaches a steady state value of K_P times the input step magnitude. The impulse response is an impulse function scaled by K_P. Both responses are sketched in Figure 1.1.

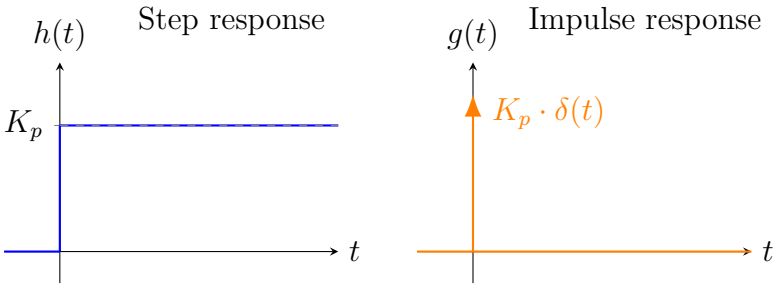


Abbildung 1.1.: P-Glied: step response $h(t) = K_P\sigma(t)$ (left) and impulse response $g(t) = K_P\,\delta(t)$ (right).

1.1.2. PT$_1$-Glied

Explanation

The PT$_1$-Glied is a first-order delay system that consists of a proportional element and a first-order part. The transfer function of a PT$_1$-Glied is given by

$$T_1 \dot{x}_a + x_a(t) = K_P x_e(t) \tag{1.11}$$

or equation 1.2, where K_P is the proportional gain and the final value of the function and T_1 is the time constant.

Parameters needed for characterization

For the characterization of a PT$_1$-Glied, the parameters needed are the proportional gain K_P and the time constant T_1. The proportional gain K_P determines the steady-state gain of the system, while the time constant T_1 determines how quickly the system responds to changes in the input. The System is also characterized by its initial tangent, which is given by $y(t) = \frac{K_P}{T_1} t$. The point at which the initial tangent intersects the steady state value of the step response can be used to determine the time constant T_1. T_1 is the time it takes for the system to reach approximately 63% of its final value in response to a step input. The threefold multiple of the time constant T_1 at which the system reaches 95% of its final value can also be used to characterize the system's response speed.

Step and Impulse response

The step response of a PT$_1$-Glied is an exponential function that approaches a steady state value of K_P times the input step magnitude, with a time constant of T_1. The impulse response is an exponential function scaled by K_P/T_1.

1.1.3. PT_2-Glied

Explanation

A PT_2-Glied is a second-order delay system that consists of a proportional element and a second-order part, which appears delayed and oscillatory behavior because of its second order differential nature. The transfer function of a PT_2-Glied is given by

$$\frac{1}{\omega_0^2}\ddot{x}_a + \frac{2D}{\omega_0}\dot{x}_a + x_a(t) = K_P x_e(t) \tag{1.12}$$

or equation 1.3 , where K_P is the proportional gain, ω_0 is the natural frequency, and D is the damping ratio.

Parameters needed for characterization

For the characterization of a PT_2-Glied, the parameters needed are the proportional gain K_P, the natural frequency ω_0, and the damping ratio D. The proportional gain K_P determines the steady-state gain of the system, while the natural frequency ω_0 and damping ratio D determine the oscillatory behavior and response speed of the system. The damping ratio D indicates whether the system is underdamped ($D < 1$), critically damped ($D = 1$), overdamped ($D > 1$) or undamped ($D = 0$).
For an underdamped system ($D < 1$), the step response will exhibit oscillations before settling to the steady state value. The natural frequency ω_0 determines the frequency of these oscillations, while the damping ratio D determines how quickly the oscillations decay. For a critically damped system ($D = 1$), the step response will reach the steady state value without oscillating, and it will do so as quickly as possible without overshooting. It is often reached by putting two PT_1-Glied in series, which results in the transfer function given by

$$T_1 T_2 \ddot{x}_a + (T_1 + T_2)\dot{x}_a + x_a(t) = K_P x_e(t) \tag{1.13}$$

or

$$h(t) = K_P \left(1 - (1 + \omega_0 t)e^{-\omega_0 t}\right) \tag{1.14}$$

For an overdamped system ($D > 1$), the step response will reach the steady state value without oscillating, but it will take longer to do so than a critically damped system. The Transfer function can be rewritten as

$$h(t) = K_P \left(1 + \frac{1}{\lambda_1 - \lambda_2}(\lambda_2 e^{-\lambda_1 t} - \lambda_1 e^{-\lambda_2 t})\right) \tag{1.15}$$

where λ_1 and λ_2 are the eigenvalues of the characteristic equation of the system, given by

$$\lambda_{1,2} = -D\omega_0 \pm \omega_0\sqrt{D^2 - 1} \tag{1.16}$$

The time constants of the system can be determined from these roots, and they indicate how quickly the system responds to changes in the input. For an undamped system ($D = 0$), the step response will exhibit sustained oscillations at the natural frequency ω_0.

For the case $0 < D < 1$, the transfer function can be rewritten as

$$h(t) = K_P \left(1 - \frac{e^{-\omega_0 D t}}{\sqrt{1-D^2}} \sin(\omega_d t + \phi)\right) \tag{1.17}$$

The Periodic time of the oscillations can be determined from the damped natural frequency

$$T_d = \frac{2\pi}{\omega_d} = \frac{2\pi}{\omega_0\sqrt{1-D^2}} \tag{1.18}$$

The transfer function can be characterized by the upper and lower envelope of the oscillations, which are given by

$$h^{\pm}(t) = K_P \left(1 \pm \frac{e^{-D\omega_0 t}}{\sqrt{1-D^2}}\right) \tag{1.19}$$

The over swing time t_{max}, which is the time at which the first maximum of the step response occurs, can be determined from the equation

$$t_{max} = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_0\sqrt{1-D^2}} \tag{1.20}$$

The overshoot h_{max}, can be determined from the equation

$$h_{max} = h(t_{max}) = K_P \left(1 + e^{-\pi D/\sqrt{1-D^2}}\right) \tag{1.21}$$

The normalized overshoot value $\ddot{u}$ can be determined from the equation

$$\ddot{u} = \frac{h_{max} - h(\infty)}{h(\infty)} = e^{\frac{-\pi D}{\sqrt{1-D^2}}} \tag{1.22}$$

which can be used to determine the damping ratio D from the overshoot of the step response. Given by the equation

$$D = \frac{|\ln(\ddot{u})|}{\sqrt{\pi^2 + \ln^2(\ddot{u})}} \tag{1.23}$$

1.1.4. I-Glied

Explanation

Integral-Glied, are transfer functions that consist of an integrator of input signal, the output $x_a(t)$ is proportional to the integral of the input $x_e(t)$, with the proportionality constant being K_I. The transfer function of an I-Glied is given by $G(s) = \frac{K_I}{s}$ 1.5. Can be rewritten as:

$$x_a(t) = K_I \int_0^t x_e(\tau) d\tau \tag{1.24}$$

Equation 1.5 can also be rewritten as

$$h(t) = K_I \int_0^t \sigma(\tau) d\tau = K_I t \tag{1.25}$$

Parameters needed for characterization

The only parameter needed to characterize an I-Glied is the integral gain K_I. This parameter determines how quickly the output will change in response to a change in the input. The step response of an I-Glied is a ramp function that increases linearly with time, with a slope of K_I times the input step magnitude. The impulse response is a step function scaled by K_I.

Step and Impulse response

The step response of an I-Glied is a ramp function that increases linearly with time, with a slope of K_I times the input step magnitude. The impulse response is a step function scaled by K_I. Both responses are sketched in Figure 1.2.

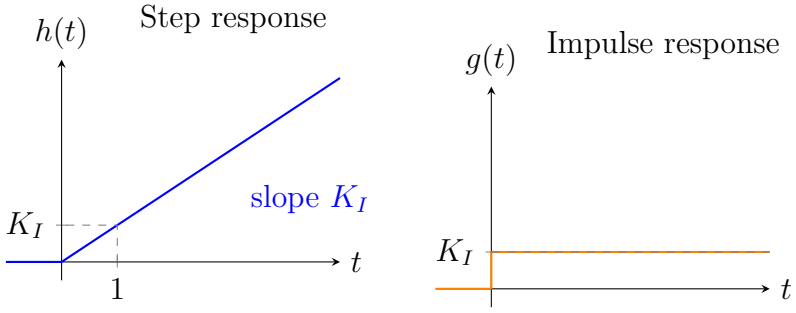


Abbildung 1.2.: I-Glied: step response $h(t) = K_I\, t$ (left) and impulse response $g(t) = K_I\, \sigma(t)$ (right).

1.1.5. IT_1-Glied

Explanation

The IT_1-Glied is a system that consists of an integrator and a first-order part. The transfer function of an IT_1-Glied is given by $G(s) = \frac{K_I}{s(T_1 s+1)}$ 1.6, where K_I is the integral gain and T_1 is the time constant. It could be seen as a combination of an I-Glied and a PT_1-Glied. The transfer function can be rewritten as

$$h(t) = K_I \left[t - T_1 \left(1 - e^{-t/T_1} \right) \right] \tag{1.26}$$

Parameters needed for characterization

For the characterization of an IT_1-Glied, the parameters needed are the integral gain K_I and the time constant T_1. The integral gain K_I determines proportionality of the function to the integral of the input, while the time constant T_1 determines how delayed the response of the system is to changes in the input.

Step and Impulse response

The step response of an IT_1-Glied is a function that increases with time, with a shape that is determined by the integral gain K_I and the delay caused by the time constant T_1. The impulse response is a function that slowly (with a delay and a shape determined by T_1) increases from zero to the value of the integral of the input impulse scaled by K_I.

2 Experiment T2: System identification

For the 3 systems investigated in this experiment, the following paragraphs will present the experimentally identified system parameters, a documentation of the results, a real-world example of such a system, and a comparison of the given system with the ideal response of the identified system. The identified systems are a PT1-glied for system 0 with a step-function input (Section 2.1), a PT2-glied for system 1 with a Dirac-impulse input (Section 2.2), and an IT1-glied for system 2 with a step-function input (Section 2.3).

2.1. System 0 — PT1

2.1.1. Experimentally Identified system parameters

The experimentally identified parameters for the PT1 system are as follows:

- Proportional gain K_P: 0.999
- Time constant T_1: 1.000 s

The corresponding identification plot, including the input step, the measured response, the analytic initial tangent and the T_1 / $3T_1$ markers, is shown in Figure 2.1.

2.1.2. Documentation of the results

Essential Commands

- *Load measurement:*

```
data0 = np.loadtxt('data/sys0.csv', delimiter=',', skiprows=1)
```

- *Detect step trigger on the input:*

```
np.where((u[:-1] < 1.0) & (u[1:] >= 1.0))
```

- *Steady-state gain:*

```
K_P = (x_inf - x_0) / (u_inf - u_0)
```

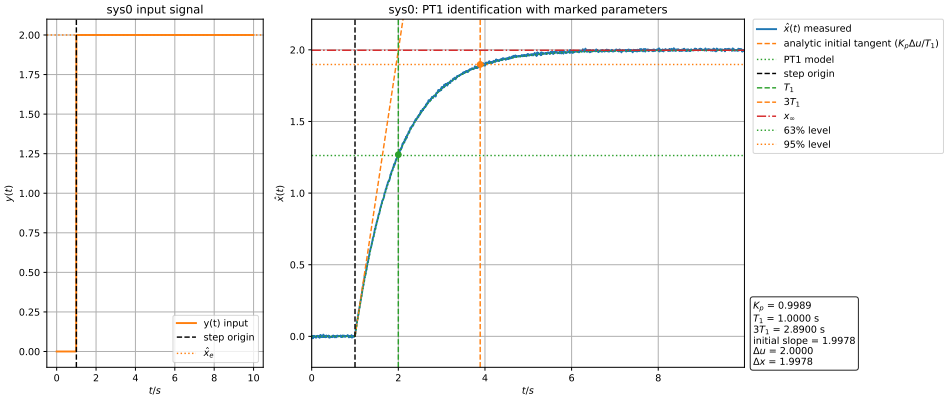


Abbildung 2.1.: System 0 (PT1): input step (left) and measured response with the analytic initial tangent $K_P \Delta u / T_1$, the PT1 model, the steady-state level x_∞ and the T_1 and $3T_1$ markers (right).

- *Time constant* T_1 from the 63 % level: first index where the normalised rise reaches $1 - e^{-1}$; cross-checked at the 95 % level ($1 - e^{-3} \Rightarrow 3T_1$):

```
rise = (x - x_0) / (x_inf - x_0)
idx_63 = np.argmax(rise >= 1 - np.exp(-1))
```

- *Initial tangent slope:*

```
slope = K_P * du / T1
```

- *PT1 step model:*

```
x_model = x_0 + dx * (1 - np.exp(-(t - t_step) / T1))
```

Differential Equation

The differential equation for a PT1 system can be derived from the transfer function and is given by

$$T_1 \dot{x}_a + x_a(t) = K_P x_e(t) \tag{2.1}$$

Transfer Function

The transfer function for a PT1 system can be derived from the differential equation and is given by

$$G(s) = \frac{K_P}{T_1 s + 1} \tag{2.2}$$

2.1.3. Realworld Example

A practical example of such a PT1 system with step excitation is a thermometer that is suddenly placed in a warm bath. The input signal $x_e(t)$ is the bath temperature, which

jumps from the ambient value to the bath value as a step at the moment the sensor is immersed. The output signal $x_a(t)$ is the temperature read by the sensor, which does not follow the bath temperature instantaneously: heat must first flow through the sensor housing into the temperature-sensitive element, so the displayed temperature rises exponentially and asymptotically approaches the bath temperature.
The proportional gain is $K_P = 1$ in this idealised case (in steady state the sensor reads exactly the bath temperature), and the time constant T_1 is set by the heat capacity of the sensor element divided by the thermal conductance between the bath and the element (housing material, wall thickness, contact area, convective heat transfer to the bath). After approximately T_1 the sensor has reached 63 % of the temperature jump, after $3T_1$ approximately 95 %. The same first-order behaviour appears in many other physical settings, for example RC low-pass filters under a step voltage or first-order mass-flow systems under a step inflow.

2.1.4. Comparison of given System with the response of the identified system

The deviation between the measured response and the ideal PT1 model is small over the whole observation window. The residual is dominated by measurement noise that is symmetric around zero and shows no systematic trend, which indicates that the identified parameters K_P and T_1 describe the system well and that no further dynamic component is missing from the model. The deviation between the identified PT1 model $h(t) = K_P\,\Delta u\,(1-e^{-t/T_1})$ and the measured response is shown in Figure 2.2; the residual against the polynomial reference fit is shown in Figure 2.3 for context.

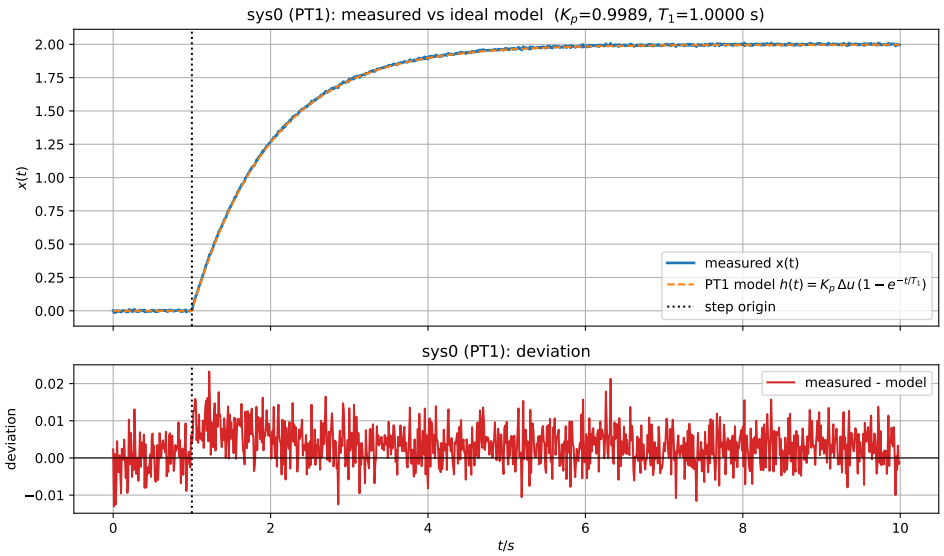


Abbildung 2.2.: System 0 (PT1): measured response together with the identified PT1 model $h(t) = K_P\,\Delta u\,(1-e^{-t/T_1})$ (top) and the resulting deviation (measured − model) versus time (bottom).

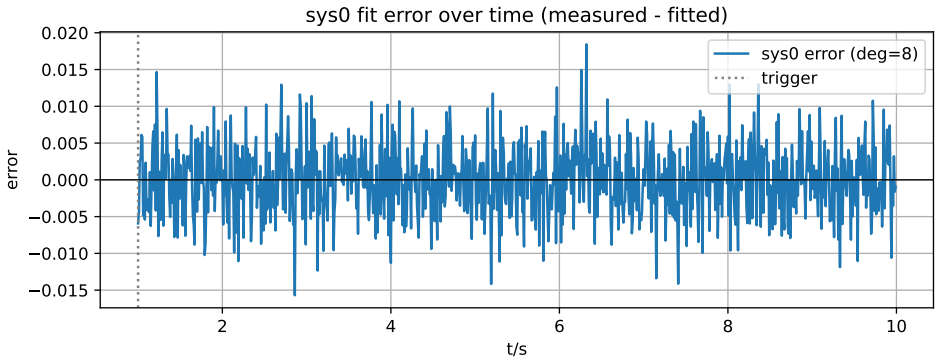


Abbildung 2.3.: System 0: residual between the measured step response and the polynomial reference fit (measured − fitted) versus time.

2.2. System 1

System 1 is a PT2 system, reacting to a Dirac impulse input and showing a damped oscillatory behaviour ($0 < D < 1$).

2.2.1. Experimentally Identified system parameters

The experimentally identified parameters for the PT2 system are as follows:

- overshoot h_{max}: 2,837
- over swing time t_{max}: 0,160 s
- the damping ratio D: 0,248
- the natural frequency ω_0: 19,69 s^{-1}
- the periodic time of the oscillations T_d: 0,320 s
- proportional gain K_P: 1,986
- input pulse area A_u: 1,000

The corresponding identification plot, including the triangular input pulse, the measured impulse response, the periodic PT2 model, the exponential envelope and the $t_{\max}$, $x_{\max}$ and T_d markers, is shown in Figure 2.4.

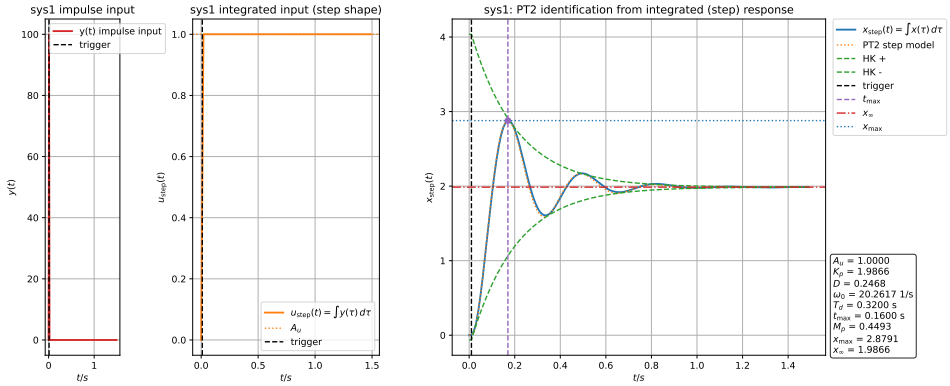


Abbildung 2.4.: System 1 (PT2): triangular input pulse (left) and measured impulse response with the underdamped PT2 model, the exponential envelope and the markers $t_{\max}$, $x_{\max}$ and T_d (right).

2.2.2. Documentation of the results

Essential Commands

- *Load measurement:*

```
data1 = np.loadtxt('data/sys1.csv', delimiter=',', skiprows=1)
```

- *Cumulative trapezoidal integration* (turns the impulse input into a step input and the impulse response into the corresponding step response):

```
cum = np.concatenate(([0.], np.cumsum(0.5 * (y[1:] + y[:-1]) *
    np.diff(t))))
```

- *Pulse area / step height:*

```
A_u = u_step[-1]
```

- *Static gain* from the integrated step response:

```
K_P = x_step_inf / A_u
```

- *First peak of the step response* ($\Rightarrow t_{\max}$, $x_{\max}$):

```
peak_idx = trigger_idx + np.argmax(x_step[trigger_idx:])
```

- *Overshoot:*

```
M_p = (x_peak - x_step_inf) / x_step_inf
```

- *Damping ratio* from the overshoot:

```
D = -np.log(M_p) / np.sqrt(np.pi**2 + np.log(M_p)**2)
```

- *Natural frequency* from $t_{\max}$, damped period T_d:

```
omega_0 = (np.pi / t_max) / np.sqrt(1 - D**2)
T_d = 2 * np.pi / (omega_0 * np.sqrt(1 - D**2))
```

- *PT2 step model* with phase φ:

```
phi = np.arctan2(np.sqrt(1 - D**2), D)
x_model = K_P * A_u * (1 - np.exp(-D * omega_0 * t) / np.sqrt(1
    - D**2) * np.sin(omega_d * t + phi))
```

Differential Equation

The differential equation for a PT2 system can be derived from the transfer function and is given by

$$\frac{1}{\omega_0^2}\ddot{x}_a + \frac{2D}{\omega_0}\dot{x}_a + x_a(t) = K_P x_e(t) \tag{2.3}$$

Transfer Function

The Transfer function for the PT2 system can be derived from the differential equation and is given by

$$G(s) = \frac{K_P}{\frac{1}{\omega_0^2}s^2 + \frac{2D}{\omega_0}s + 1} \tag{2.4}$$

2.2.3. Realworld Example

A classic example of a PT2 system with underdamped behaviour is a mass–spring–damper system excited by a step force. The input signal $x_e(t)$ is the applied force $F(t)$, which jumps from zero to a constant value at the moment of excitation. The output signal $x_a(t)$ is the displacement of the mass from its rest position, which does not jump immediately to the new equilibrium: the mass first overshoots, then performs damped oscillations around the steady-state value before settling.
The equation of motion $m\,\ddot{x}_a + c\,\dot{x}_a + k\,x_a = F(t)$ matches the standard PT2 form with proportional gain $K_P = 1/k$ (the steady-state compliance, i.e. the inverse of the spring stiffness), natural frequency $\omega_0 = \sqrt{k/m}$ (set by mass m and stiffness k), and damping ratio $D = c/(2\sqrt{mk})$ (set by the damping coefficient c relative to the critical damping). For $0 < D < 1$ the response shows the characteristic overshoot and decaying oscillation, with a damped period $T_d = 2\pi/(\omega_0\sqrt{1-D^2})$ and a first peak at $t_{\max} = \pi/(\omega_0\sqrt{1-D^2})$. The same second-order behaviour appears in many other physical settings, for example a series RLC circuit driven by a step voltage (output: capacitor voltage, with $\omega_0 = 1/\sqrt{LC}$ and $D = (R/2)\sqrt{C/L}$), a moving-coil galvanometer responding to a step current, or a passive vehicle suspension reacting to a sudden load change.

2.2.4. Comparison of given System with the response of the identified system

The measured response and the ideal PT2 model agree well over most of the observation window, with the residual after the first oscillation dominated by measurement noise. Larger deviations are visible at the very beginning of the response: the excitation in the experiment is not an ideal Dirac impulse but a short triangular pulse of finite width, so the measured signal rises more gradually than the ideal impulse response and the first peak amplitude differs slightly from the model prediction. Once the impulse has passed, the damped oscillation matches the identified D, ω_0 and K_P within the noise level. The deviation between the integrated step response $\int x(\tau)\,d\tau$ and the identified PT2 model is shown in Figure 2.5; the residual against the polynomial reference fit of the impulse response is shown in Figure 2.6 for context.

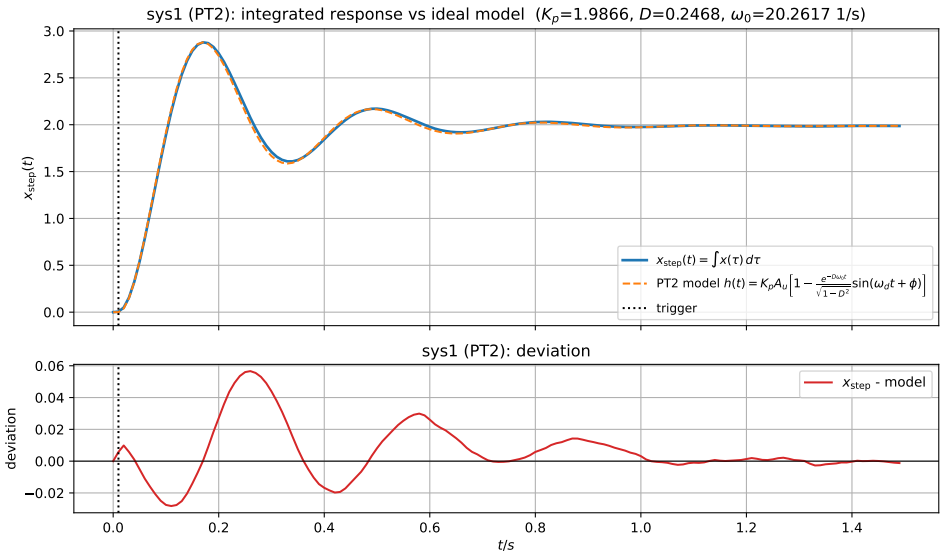


Abbildung 2.5.: System 1 (PT2): integrated step response $\int x(\tau)\,d\tau$ together with the identified PT2 model $h(t) = K_P\,A_u\left[1 - \frac{e^{-D\omega_0 t}}{\sqrt{1-D^2}}\sin(\omega_d t + \phi)\right]$ (top) and the resulting deviation versus time (bottom).

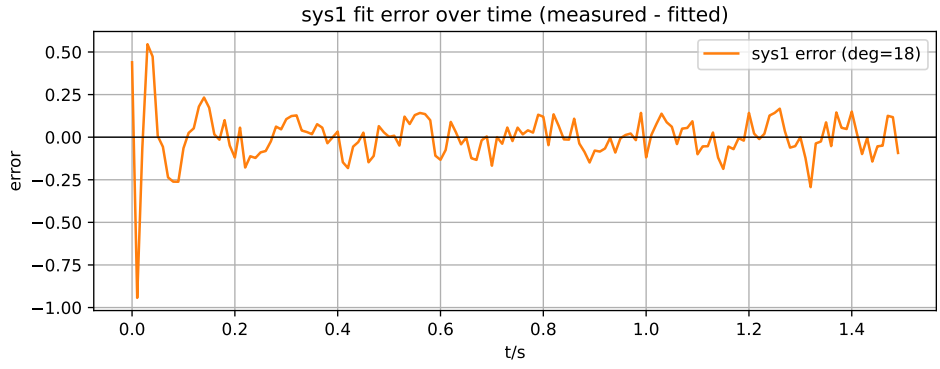


Abbildung 2.6.: System 1: residual between the measured impulse response and the polynomial reference fit (measured − fitted) versus time.

2.3. System 2

2.3.1. Experimentally Identified system parameters

The experimentally identified parameters for the IT1 system are as follows:

- Integral gain K_I: 0,300 s^{-1}
- Proportionalbeiwert K_P: 10 (given)
- Time constant T_1: 0,493 s

The corresponding identification plot, including the input step, the measured response, the IT1 model, the pure-I reference $K_P\,K_I \int u\,dt$ and the asymptotic tangent with its time-axis intersection, is shown in Figure 2.7.

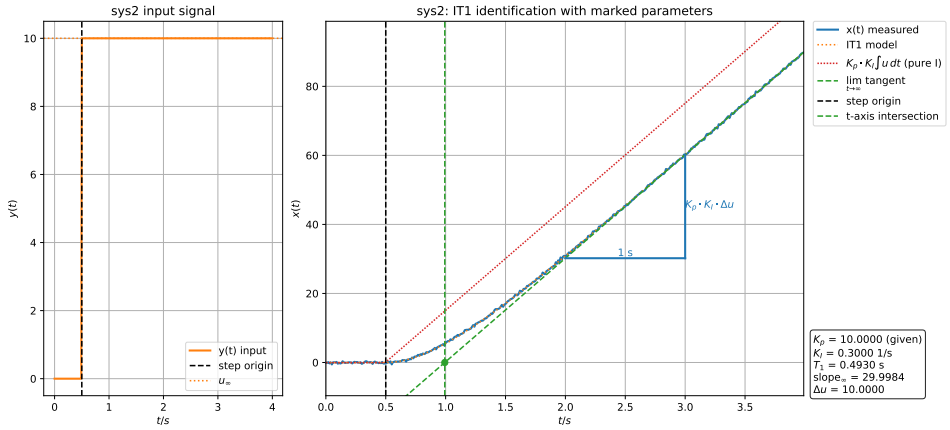


Abbildung 2.7.: System 2 (IT1): input step (left) and measured response with the IT1 model, the pure-I reference $K_P\,K_I \int u\,dt$ and the asymptotic tangent with its time-axis intersection at $t = t_{\mathrm{step}} + T_1$ (right).

2.3.2. Documentation of the results

Essential Commands

- *Load measurement:*

```
data2 = np.loadtxt('data/sys2.csv', delimiter=',', skiprows=1)
```

- *Detect step trigger on the input:*

```
np.where((u[:-1] < 1.0) & (u[1:] >= 1.0))
```

- *Linear fit on the asymptotic tail* (last $\sim 25\,\%$ of samples):

```
slope_inf, intercept_inf = np.polyfit(t[tail:], x[tail:], 1)
```

- *Integral gain* from the asymptotic slope and the given $K_P = 10$:

```
K_I = slope_inf / (du * K_P_plant)
```

- *Time-axis intersection* of the asymptote:

```
t_intersect = -intercept_inf / slope_inf
```

- *Time constant* from the intersection:

```
T_1 = t_intersect - t_step
```

- *IT1 step model:*

```
x_model = x_0 + K_P * K_I * du * (tau - T_1 * (1 - np.exp(-tau /
    T_1)))
```

- *Pure-I reference* for visual comparison: cumulative trapezoidal of $u(t) - u_0$ scaled by $K_P \cdot K_I$.

Differential Equation

The differential equation for an IT1 system can be derived from the transfer function and is given by

$$T_1 \dot{x}_a + x_a(t) = K_I \int x_e(t) dt \tag{2.5}$$

Transfer Function

The transfer function for an IT1 system can be derived from the differential equation and is given by

$$G(s) = \frac{K_I}{s(T_1 s + 1)} \tag{2.6}$$

2.3.3. Realworld Example

A clear example of an IT1 system is a water tank fed by an inflow $x_e(t)$ and drained through a linear resistance to a downstream sink. The inflow corresponds to the input signal, and the water level $x_a(t)$ to the output signal. The storage of the tank gives the system its integrative behaviour — as long as the inflow exceeds the outflow, the level keeps rising — while the proportional outflow causes a first-order lag with time constant $T_1 = A/k$, where A is the cross-sectional area of the tank and k is the linear hydraulic conductance of the outflow.
The integral gain is $K_I = 1/A$ (a unit inflow integrates into a unit volume change, divided by the cross-section to obtain a level change). After a step change in inflow the level first lags behind for a duration of order T_1 and then approaches a straight ramp of slope $K_I \, \Delta x_e$. The same IT1 behaviour appears in other accumulating systems with first-order losses, for example a heated body with a fixed heat input and a proportional

convective loss to the ambient, or a capacitor charged through a current source while leaking through a parallel resistor.

2.3.4. Comparison of given System with the response of the identified system

The measured response and the ideal IT1 model agree closely over the whole observation window. The remaining differences are dominated by measurement noise that is symmetric around zero and does not show a systematic trend, so the identified K_I and T_1 describe the system well and no additional dynamic effect needs to be added to the model. The deviation between the identified IT1 model $h(t) = K_P\, K_I\, \Delta u\, [t - T_1(1 - e^{-t/T_1})]$ and the measured response is shown in Figure 2.8; the residual against the polynomial reference fit is shown in Figure 2.9 for context.

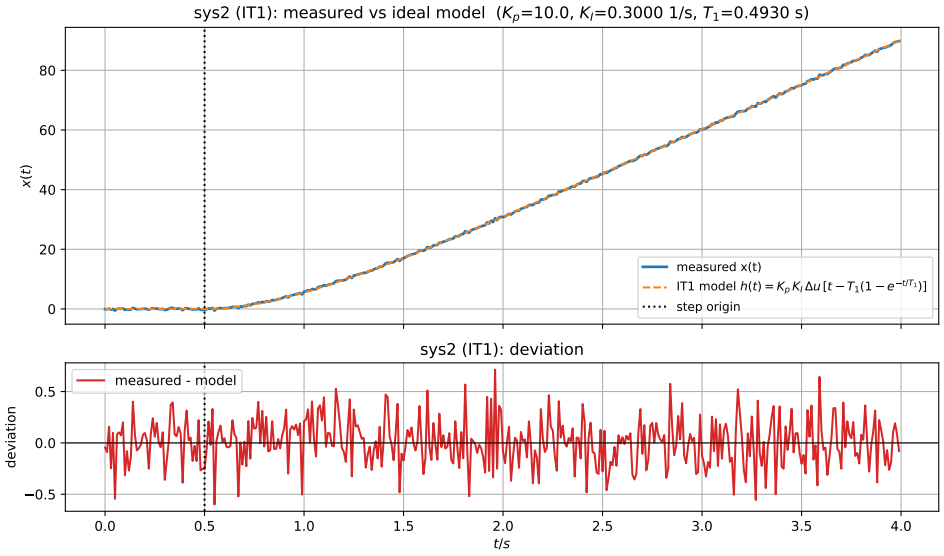


Abbildung 2.8.: System 2 (IT1): measured response together with the identified IT1 model $h(t) = K_P\, K_I\, \Delta u\, [t - T_1(1 - e^{-t/T_1})]$ (top) and the resulting deviation versus time (bottom).

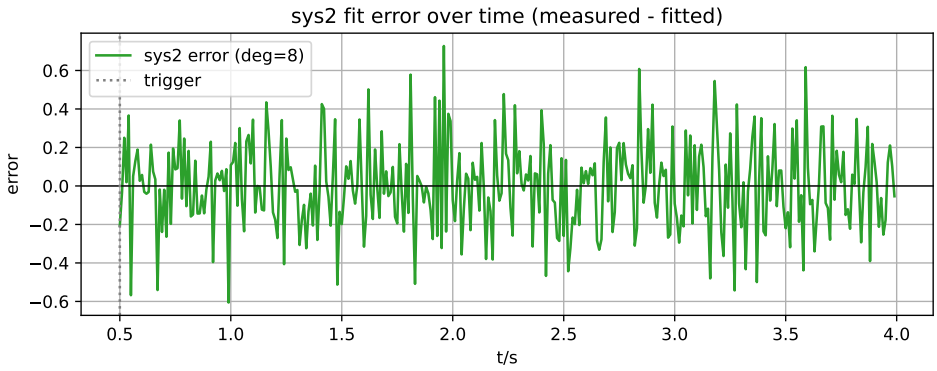


Abbildung 2.9.: System 2: residual between the measured step response and the polynomial reference fit (measured − fitted) versus time.

A Appendix Frontmatter

A.1. Declarations

1. Wir erkläre hiermit, dass wir dieses Dokument selbstständig verfasst, noch nicht anderweitig für Prüfungszwecke vorgelegt, keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie wörtliche und sinngemäße Zitate als solche gekennzeichnet haben.

Regensburg, den 2026-04-29,
Marius Off *Matrikelnummer* : 3416002,
Janis Graf *Matrikelnummer* : 3377251
Ort, Datum und Unterschrift

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B Appendix Code and Resources

B.1. Code used for the analysis of the Preperations and the Experiment

Editorial note — Source code omitted

The full source code for the analysis (a Jupyter notebook of approximately 2,000 lines) is *not* included in this appendix. Including it as a verbatim listing caused build instability and was therefore deliberately omitted.
The complete source can be obtained by either of the following means:

- By email request to `janis.graf@st.oth-regensburg.de`
- Via requesting access to the GitHub repository: `https://github.com/janisgra/RTP-T2`